Brain Tumar Classification

Name: Madonna Ashraf Fakhry

Section: AI2

ID: 2023016838

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Introduction

One of the most serious diseases of the human brain is brain tumors. The earlier the diagnosis, the better the prospects for a successful treatment and patient survival. Brain tumor detection in MRI imaging is a common practice but manual analysis can be difficult and time-consuming.

Deep Learning techniques and in particular Convolutional Neural Networks (CNNs) have shown promising performance in classifying medical images. In this project we built a model based on DenseNet for classification of brain MRI images into four classes:

- Glioma
- Meningioma
- Pituitary Tumor
- No Tumor

The goal of this project is to develop a precise model to assist in the automated classification of brain tumors.

Method

Data preparation

The MRI images were resized to 224 × 224 pixels and normalized to training. The dataset was split into:

- Training data set
- Validation data set
- Testing data set

Images were converted into tensors with PyTorch to be ready for model to train and validate and finally test.

Model Architecture

The classification was done using a DenseNet based CNN model. The architecture consists of:

- Convolutional layers
- Dense blocks
- Pooling layers
- Fully connected layers

Dropout layers and ReLU activation were used to improve performance and reduce overfitting.

Education & Training

The model was trained for several epochs with Cross Entropy Loss. We monitored the training and validation accuracy and loss during the training and testing.

Dataset Description

The dataset is a collection of Brain MRI images gathered from Kaggle. Images are categorized in the following four categories:

- Glioma
- Meningioma
- Pituitary
- No Tumor

The dataset was split into training and testing folders already.

Dataset Split	Purpose
Training	Model training
Testing	Model evaluating and mentoring

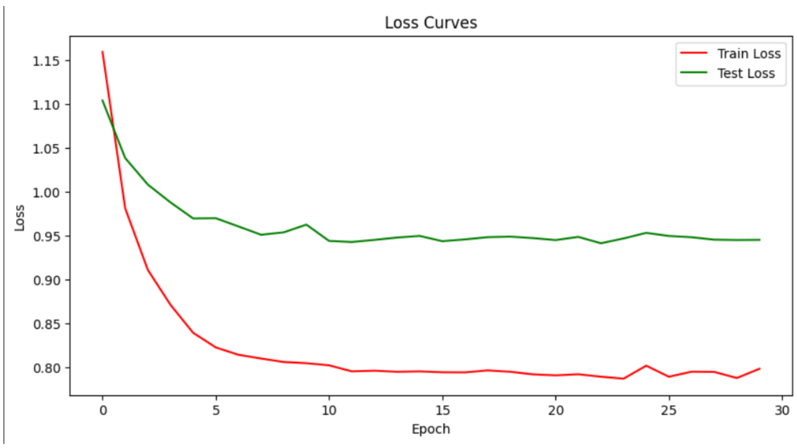
You can visit dataset from this link:

https://www.kaggle.com/datasets/masoudnickparvar/brain-tumor-mri-dataset

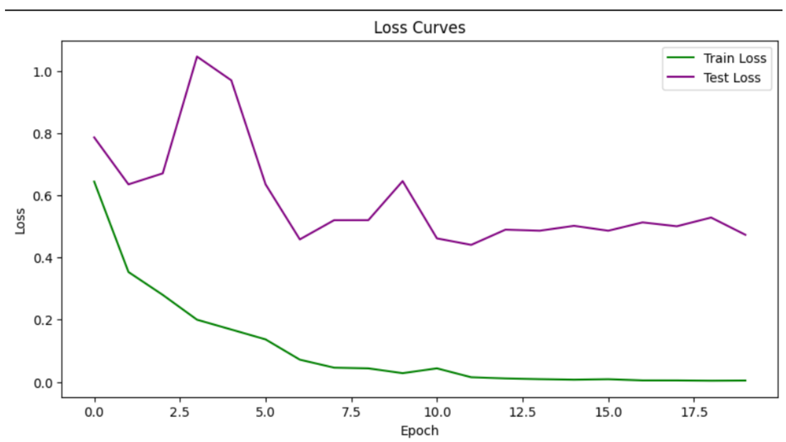
Experiment

Experiment 1	Experiment 2
Simple Dense Net	Dense Net121
Optimizer: Adam	Optimizer: Adam
Learning Rate: 0.0001	Learning Rate: 1e-4
Epoches:30	Epoches:20

The loss curves describe the learning behavior of the model during the training . In both experiments, the training loss decreased continuously, demonstrating that the model successfully learned important features from the MRI images.



In the first experiment, the validation/test loss gradually decreased and was stabilized relatively, which means better generalization and more stable learning performance.



In the second experiment, the training loss decreased very quickly and was close to zero, whereas the validation/test loss fluctuated considerably. This means that the model may have been slightly overfitted, having learned the training data really well, but generalizing less well on unseen data.

In summary, the first experiment performed more stably, whereas the second experiment learned more quickly but produced less stable results on the validation set.

Results

The model was able to classify brain MRI images with a good performance!

Metrics Used

- Accuracy
- Loss
- Precision
- Recall

	Experiment 1	Experiment 2
Accuracy	63.25%	93.38%
Loss	0.9454	0.4729
Precision	glioma 0.53 meningioma 0.54 notumor 0.69 pituitary 0.72	glioma 0.98 meningioma 0.89 notumor 0.91 pituitary 0.96
Recall	glioma 0.62 meningioma 0.24 notumor 0.86 pituitary 0.80	glioma 0.80 meningioma 0.95 notumor 1.00 pituitary 0.99

Discussion

Results indicate that DenseNet is effective in brain tumor classification. Adam optimizer performed the best accuracy .

The model performed well for most of the classes especially for pituitary and no-tumor images. Some confusion was observed between glioma and meningioma for similar image features.

Conclusion

This project proposed Deep Learning model for brain tumours classification using MRI images. The DenseNet-based model showed high accuracy and the efficiency of Deep Learning in medical image classification.

Future Work

Possible future improvements include:

- Using larger datasets
- Using pretrained models
- Deploying the model as a web application